Supervised Machine Learning
Linear Regression

CSE4107: Artificial Intelligence

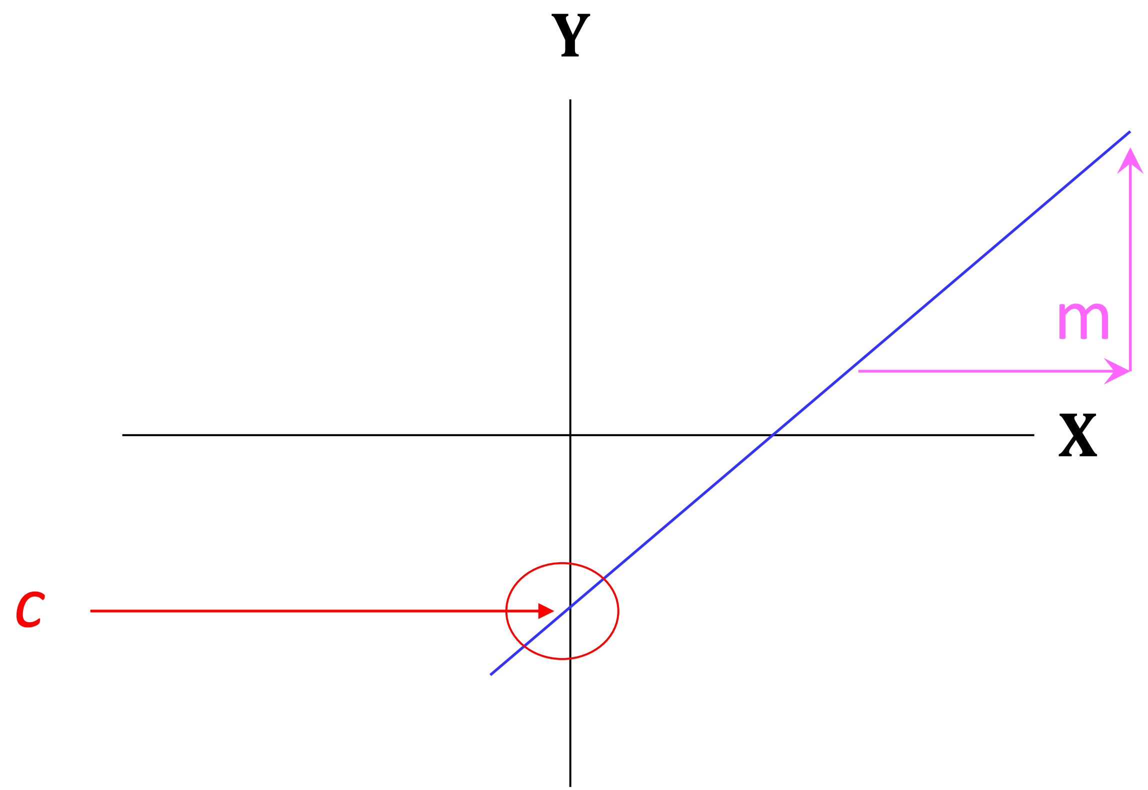
References

- Medium
- Real Python
- towards data science

What is Linear?

$$\boldsymbol{y = mx + c}$$

A slope of 2 means that every 1-unit change in X yields a 2-unit change in Y.



What is Regression?

- **Regression** is a statistical procedure that determines the equation for the straight line that best fits a specific set of data.
- It is a statistical method used in various sectors that attempts to determine the strength and character of the relationship between one dependent variable (usually denoted by Y) and a series of other variables (known as independent variables).

Linear Regression

- How well a set of data points fits a straight line can be measured by calculating the distance between the data points and the line.
- The total error between the data points and the line is obtained by squaring each distance and then summing the squared values.
- The regression equation is designed to produce the minimum sum of squared errors.

Prediction and Regression Equation

If you know something about X, this knowledge helps you predict something about Y. (Sound familiar?…sound like conditional probabilities?)

Regression Equation

$$\boldsymbol{y} = \boldsymbol{mx} + \boldsymbol{c} \longrightarrow \boldsymbol{E}(\boldsymbol{y_i}|\boldsymbol{x_i}) = \boldsymbol{mx} + \boldsymbol{c} \longrightarrow \boldsymbol{E}(\boldsymbol{y_i}|\boldsymbol{x_i}) = \beta\boldsymbol{x_i} + \alpha$$

Regression Line and Equation

- We will write an estimated regression line based on sample data as

$$\hat{y} = b_0 + b_1 x$$

- The method of least squares chooses the values for b_0, and b_1 to minimize the sum of squared errors

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} \left(y - b_0 - b_1 x\right)^2$$

Regression Line and Equation

- Using Calculus, we obtain the following formula:

$$b_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{n\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n\sum_{i=1}^{n} x_i^2 - (\sum_{i=1}^{n} x_i)^2}$$

Or, $b_1 = r\frac{S_y}{S_x}$ and $b_0 = \bar{y} - b_1\bar{x}$

Example Math

Fitted regression line can be used to estimate the mean value of y for a given value of x.

Example

The weekly advertising expenditure (x) and weekly sales (y) are presented in the following table.

y	x
1250	41
1380	54
1425	63
1425	54
1450	48
1300	46
1400	62
1510	61
1575	64
1650	71

Example Math

From previous table we have:

$$n = 10 \qquad \sum x = 564 \qquad \sum x^2 = 32604$$

$$\sum y = 14365 \qquad \sum xy = 818755$$

The least squares estimates of the regression coefficients are:

$$b_1 = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2} = \frac{10(818755) - (564)(14365)}{10(32604) - (564)^2} = 10.8$$

$$b_0 = 1436.5 - 10.8(56.4) = 828$$

y	x
1250	41
1380	54
1425	63
1425	54
1450	48
1300	46
1400	62
1510	61
1575	64
1650	71

Example Math

- The estimated regression function is:

$$\hat{y} = 828 + 10.8x$$

- This means that if the weekly advertising expenditure is increased by $1 we would expect the weekly sales to increase by $10.8.
- Fitted values for the sample data are obtained by substituting the x value into the estimated regression function.
- For example if the advertising expenditure is $50, then the estimated Sales is:
- This is called the point estimate (forecast) of the mean response (sales).

y	x
1250	41
1380	54
1425	63
1425	54
1450	48
1300	46
1400	62
1510	61
1575	64
1650	71

Regression Equation

$$E(y_i|x_i) = \underbrace{\beta x_i + \alpha}_{\text{Fixed – exactly on the line}} + random\ error_i$$

Follows a Normal Distribution

For Multiple predictor:

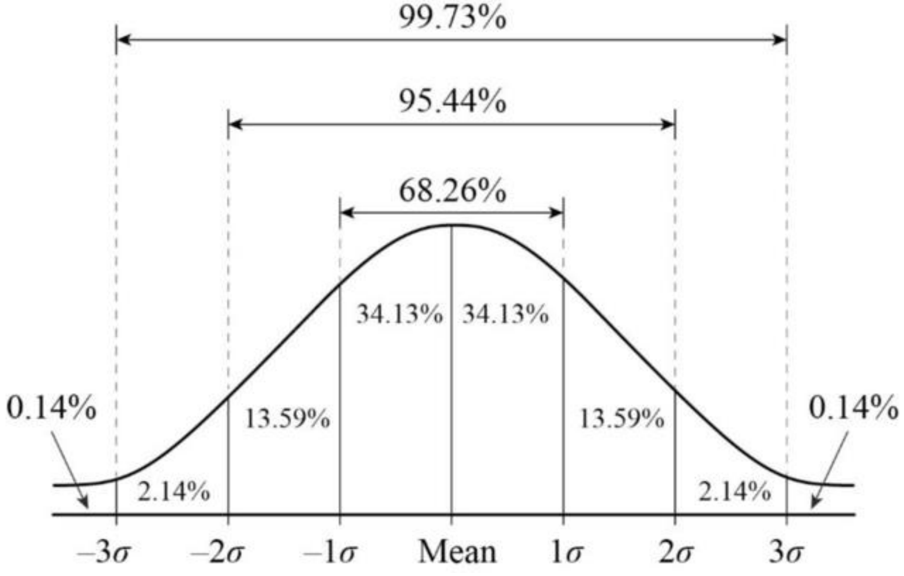
$$E(y) = \alpha + \beta_1 * X + \beta_2 * W + \beta_3 * Z \ldots$$

Assumption

Linear regression assumes that…

1. The relationship between X and Y is **linear**
2. Y is **distributed normally** at each value of X
3. The **variance** of Y at every value of X is the same (homogeneity of variances)
4. The **observations (x)** are independent

Normal Distribution



Example

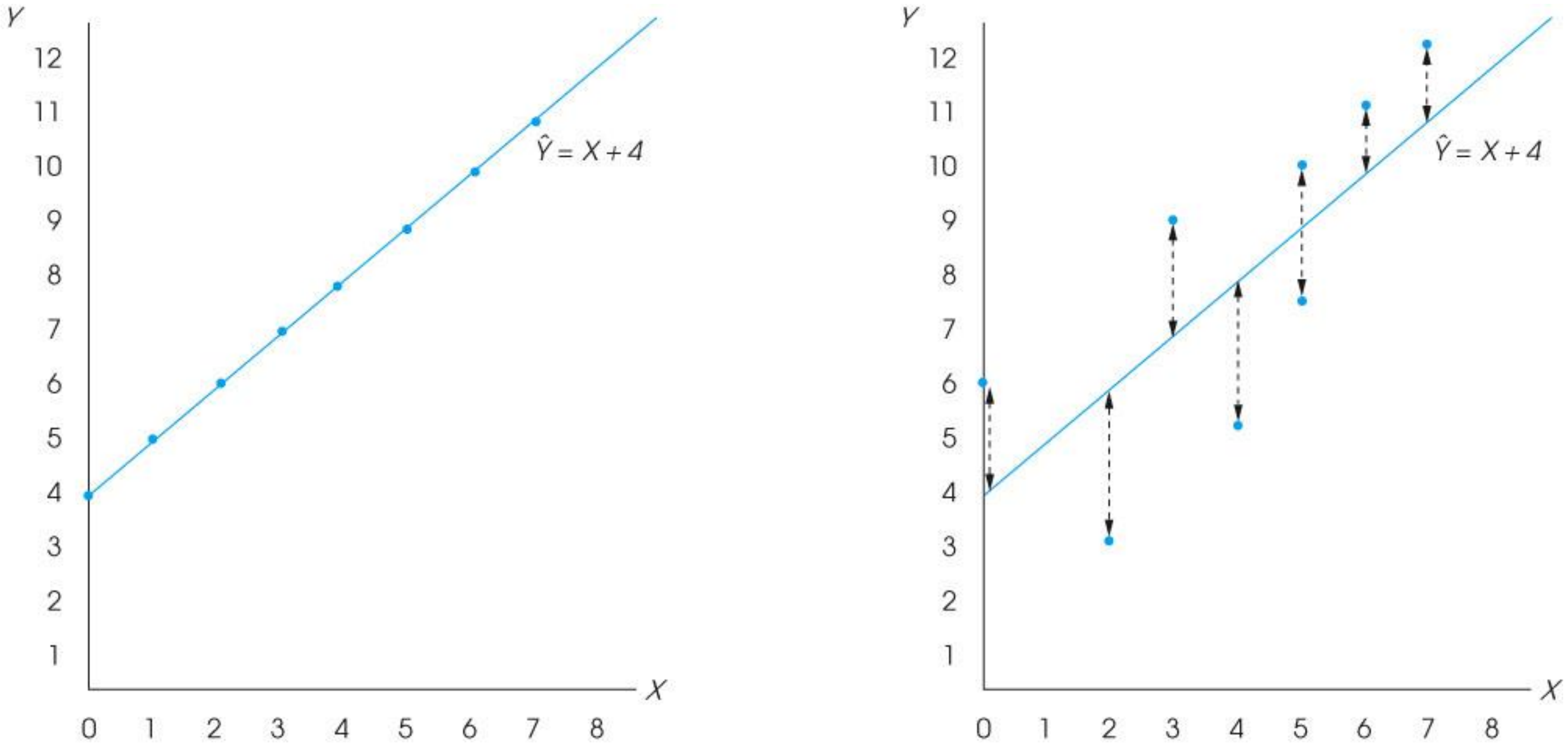


Fig. 1 A scatter plot showing data points that perfectly fit the regression line defined by the equation $Y = X + 4$.

Finding the best-fitting line

Question: Find the best-fitting line for the data points **(1, 2), (2, 3), (4, 7), (5, 5), (7, 11).**

To find the line $y = mx + c$ of best fit through these five points, the goal is to minimize the **sum of the squares** of the differences between the *y*-coordinates and the predicted *y*-coordinates based on the line and the *x*-coordinates. This is—

For $(1, 2) \rightarrow 2 = 1.m + c \quad or, \ \ (1.m\ + c) - 2 = 0$

$$\boldsymbol{(1.m\ + c - 2)^2 + (2.m\ + c - 3)^2 + (4.m\ + c - 7)^2 + (5.m\ + c - 5)^2 + (7.m\ + c - 11)^2}$$

Finding the best-fitting line

Question: Find the best-fitting line for the data points **(1, 2), (2, 3), (4, 7), (5, 5), (7, 11).**

$$(\mathbf{1.m + c - 2})^2 + (\mathbf{2.m + c - 3})^2 + (\mathbf{4.m + c - 7})^2 + (\mathbf{5.m + c - 5})^2 + (\mathbf{7.m + c - 11})^2$$

This is a quadratic polynomial in m*m* and *b*, and is minimized by taking partial derivatives with respect to *m*, using the chain rule, and setting them equal to 0. This gives

$$2(1.m + c - 2) + 2.2(2.m + c - 3) + 2.4(4.m + c - 7) + 2.5(5.m + c - 5) + 2.7(7.m + c - 11) = 0$$
$$or, 2(1.m + c - 2) + 4(2.m + c - 3) + 8(4.m + c - 7) + 10(5.m + c - 5) + 14(7.m + c - 11) = 0$$
$$or, (1.m + c - 2) + 2(2.m + c - 3) + 4(4.m + c - 7) + 5(5.m + c - 5) + 7(7.m + c - 11) = 0$$

which reduces to,

$$95m + 19b = 138$$
$$or, \mathbf{19m + 5b = 28}$$

Finding the best-fitting line

Question: Find the best-fitting line for the data points **(1, 2), (2, 3), (4, 7), (5, 5), (7, 11).**

$$\mathbf{19}\boldsymbol{m} + \mathbf{5}\boldsymbol{b} = \mathbf{28}$$

$$or, 5b = -19m + 28$$

$$or, b = -\frac{19}{5}m + \frac{28}{5}$$

Plot it on the Graph (the line and the points)

Miscellaneous

- Pros and Cons
- Residual error